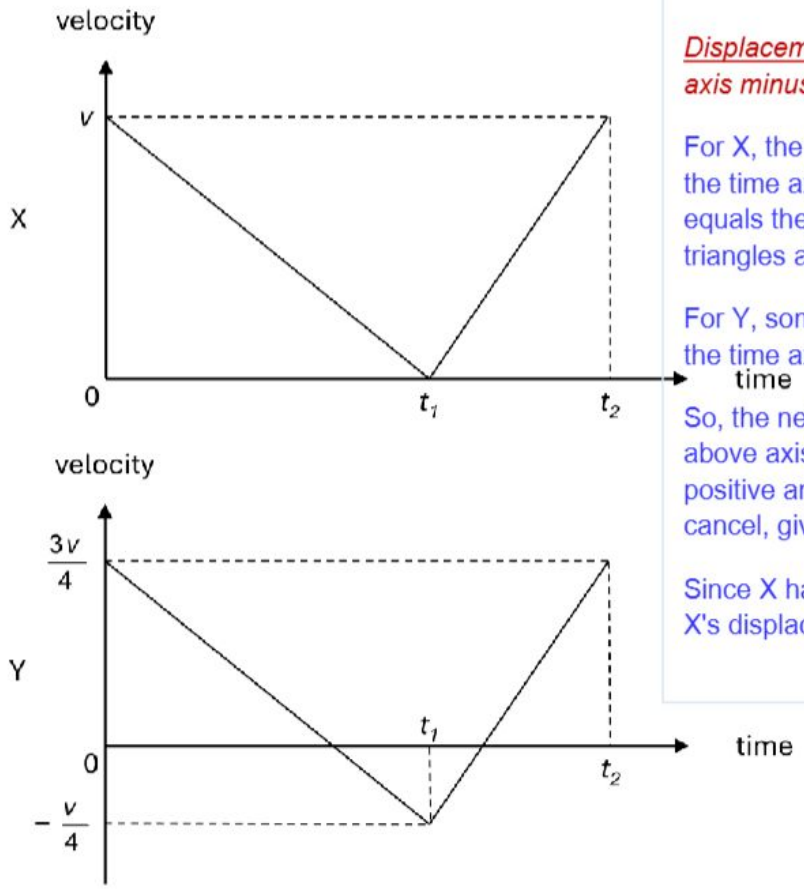
2 The motions of two objects X and Y along a straight line are shown in the following velocity-time graphs.



On a velocity-time graph:

Displacement = area above the time axis minus area below the time axis

For X, the entire graph sits above the time axis, so the displacement equals the total area of the two triangles above the axis.

For Y, some of the graph is below the time axis.

So, the net displacement = (area above axis) − (area below axis). The positive and negative areas partially cancel, giving a smaller net result.

Since X has no cancellation at all, X's displacement is larger.

How do the magnitudes of the displacement and the total distance travelled compare for the two objects from time 0 to time t_2?

Ans: A

	displacement	**distance**

A	different	different
B	different	same
C	same	different
D	same	same

below| (all areas treated as positive)

For X, since the graph never goes negative, total distance = displacement = same total area as before.

For Y, you'd add the area below the axis back in (instead of subtracting it). However, the key insight is that Y's graph reaches a lower peak on the positive side (only $3v/4$ compared to X's v) and the negative dip "wastes" some of the area on cancellation. After adding up, the total area for Y still comes out smaller than X's.

3 The force-extension graph of a spring is as shown.